TARANEH SAADATI

RESEARCH FELLOW | DATA SCIENTIST | RESILIENCE & FORECASTING ANALYTICS
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PROFESSIONAL PROFILE

Taraneh Saadati is a Postdoctoral Research Fellow at Brunel Business School, Brunel University London. Her PhD is in Energy Science and Technology, with a focus on data science, and she has an interdisciplinary background in mathematics, industrial engineering, and socio-technical systems. She is experienced in designing Python-based analytics and AI-driven decision-support tools for risk assessment, supply-chain resilience and operational planning. Her research expertise spans machine learning, deep learning, ensemble modelling, hierarchical and probabilistic time-series forecasting, explainable AI and renewable-energy analytics. She has published in peer-reviewed journals and has experience in collaborative, internationally funded research with practical applications in sustainability.

CORE SKILLS

Resilience & Risk Analytics	Supply-chain resilience; risk assessment; risk scoring and mathematical modeling; project design; business analysis; disruption monitoring; software development
AI & Data Science	Machine learning; deep learning; ensemble learning; explainable AI; feature engineering; model optimisation; data preprocessing; validation and performance evaluation
Forecasting & Operations Research	Time-series forecasting; hierarchical forecasting; forecast reconciliation; probabilistic forecasting; rolling-origin evaluation; optimisation; multi-criteria decision analysis
Technical	Python; pandas; NumPy; scikit-learn; TensorFlow/Keras; PyTorch; XGBoost; statistical analysis; data visualisation; Git/GitHub

PROFESSIONAL & RESEARCH EXPERIENCE

Research Fellow **May 2026 - Present**
Brunel University London - Horizon Europe NARRATE Project | London, UK

- Contribute to the development of a Resilient Digital Module (RDM) supporting risk assessment, resilience monitoring and decision-making across complex supply chains and manufacturing networks.
- Design and implement Python-based resilience analytics, including risk scoring, impact assessment, propagation analysis, time-to-recovery estimation, trend and drift analysis, and decision-support reporting.
- Translate research requirements into practical analytical workflows and user-facing tools that help organisations anticipate, monitor, prioritise and respond to disruptions.
- Collaborate with academic researchers and European industry partners on an applied research programme focused on resilience, sustainability and real-world impact.

PhD Researcher **Feb 2019 - May 2025**
Istanbul Technical University (ITU), Energy Institute | Istanbul, Türkiye

- Developed and evaluated machine-learning, deep-learning, ensemble and chaotic models for photovoltaic energy forecasting using real-world, high-frequency solar plant data.
- Implemented end-to-end Python workflows covering preprocessing, outlier handling, feature engineering, hyperparameter optimisation, validation, performance comparison and visualisation.

- Designed a deep-learning stacking meta-model and investigated nonlinear phase-space reconstruction and Echo State Networks for improved forecasting accuracy.

PhD Visiting Researcher **Oct 2023 - Feb 2025**

Cardiff University / Cardiff Business School | Cardiff, UK

- Conducted research on hierarchical and probabilistic time-series forecasting for NHS emergency admissions, comparing statistical and machine-learning models with forecast-reconciliation methods.
- Collaborated with forecasting researchers on empirical design, rolling evaluation, interpretation and academic publication.

Research Assistant **Dec 2023 - Mar 2024**

Cardiff Business School | Cardiff, UK

- Supported applied forecasting research, data preparation, modelling and analysis within a collaborative academic environment.

Research Scholar **Jan 2023 - Mar 2024**

TÜBİTAK-funded Research Project | Istanbul, Türkiye

- Developed research for a funded project on a sensitivity-analysis-based explainable AI method and its application to deep-learning models for remote-sensing imagery.

Data Scientist **Aug 2021 - Feb 2022**

Presify / Inavitas | Türkiye

- Analysed energy-consumption data and developed time-series forecasting models for Energy Management System R&D.

Teaching Assistant - Mathematical Logic **2014 - 2015**

University of Tehran, Faculty of Mathematics and Science | Tehran, Iran

- Supported teaching and learning activities in mathematical logic.

EDUCATION

PhD, Energy Science and Technology 2019 - 2025

Istanbul Technical University, Energy Institute | Thesis: Forecasting of Produced Output Electricity in Photovoltaic Power Plants

MSc, Industrial Engineering - Socio-Economic Systems 2016 - 2018

Kharazmi University

BSc, Applied Mathematics 2010 - 2015

University of Tehran

SELECTED RESEARCH PROJECTS

- Resilient Digital Module (RDM) for supply-chain and manufacturing resilience - Horizon Europe NARRATE: risk assessment, RDM development, recovery and decision-support analytics.
- Forecasting of Produced Output Electricity in Photovoltaic Power Plants – PhD Thesis Research Grant, Istanbul Technical University, Scientific Research Projects (BAP) Project (MDK-2022-44133).
- Hierarchical forecasting for national health-service emergency admissions: ARIMA, ETS and XGBoost with Bottom-Up, MinT and ERM reconciliation; point and probabilistic evaluation.
- Sensitivity-based explainable AI for remote-sensing scene classification within a TÜBİTAK-funded research project.
- Environmental economic dispatch on the IEEE 30-bus, six-generator system using decomposition and evolutionary multi-objective optimisation.
- Energy-system analysis for smart homes (Tableau); solar-irradiance forecasting with neural networks (Python); CSP feasibility analysis (SAM); building-energy improvement analysis (eQUEST).

PUBLICATIONS & RESEARCH OUTPUTS

- Saadati, T., & Barutcu, B. (2025). Forecasting Solar Energy: Leveraging Artificial Intelligence and Machine Learning for Sustainable Energy Solutions. Journal of Economic Surveys, 39(5), 1929-1946. https://doi.org/10.1111/joes.12678

- Saadati, T., & Barutcu, B. (2025). Optimized Solar Energy Forecasting for Sustainable Development Using Machine Learning, Deep Learning, and Chaotic Models. International Journal of Energy Economics and Policy, 15(1), 110-120. https://doi.org/10.32479/ijeep.17766

- Saadati, T., & Taskin, G. (2024). A Sensitivity-Based Explainable Method for Remote Sensing Scene Classification. 2024 IEEE International Geoscience and Remote Sensing Symposium (IGARSS), 3026-3029. https://doi.org/10.1109/IGARSS53475.2024.10641899

- Saadati, T., & Barutcu, B. Enhancing Solar Energy Forecasting through Chaos Theory and Echo State Networks: A Case Study of the Konya Eregli Solar Power Plant. Conference contribution, ICEESEN 2024.

- Saadati, T., Rostami-Tabar, B., & Heravi, S. Hierarchical Time Series Forecasting in National Health Service. Manuscript under review.

- Saadati, T. Solar Energy Production Time Series Data. Mendeley Data, Version 3. https://doi.org/10.17632/2tpv28kr83.3

SELECTED CONFERENCE PAPERS

- Nazemi, A., Saadati, T., & Yarmohammadi, M. (2018). A Review and Evaluation in Energy Intensity Index. International Congress on Science and Engineering, Hamburg, Germany.

- Nazemi, A., Saadati, T., & Yarmohammadi, M. (2018). Multi-objective Modeling of Economic and Environmental Dispatching Based on Weighted Sum Approach. International Congress on Science and Engineering, Hamburg, Germany.

- Nazemi, A., Saadati, T., & Yarmohammadi, M. (2017). Evaluation of Different Energy Sources in Greece's Road Transportation Sector Using Analytic Hierarchy Process. 3rd International Conference on Industrial Engineering and Management, Tehran, Iran.

LICENCES & CERTIFICATIONS

- Deep Learning with PyTorch
- Winter school training in solar radiation forecasting at Research Center O.I.E. of MINES Paris – PSL (France)
- Educational Course in Photovoltaic Power Plants

LANGUAGES

Persian - Native or bilingual | **English** - Full professional | **Turkish** - Full professional